

1) Given the function $f(x) = -2x^2 + 3x + 5$ on the interval $[0, 4]$, find the value c guaranteed by the Mean Value Theorem.

$$\frac{f(4) - f(0)}{4 - 0}$$

$$\frac{-15 - 5}{4 - 0} = \frac{-20}{4} = -5 \quad (\text{Average Slope})$$

Find the derivative and set it to the average slope

$$f'(x) = -4x + 3$$

$$-4c + 3 = -5$$

$$-4c = -8$$

$$c = 2$$

$$f(4) = -2(4)^2 + 3(4) + 5$$

$$= -32 + 12 + 5$$

$$= -15$$

$$f(0) = 5$$

2) A rectangular pigpen is to be constructed with material that cost \$10 per foot on 3 sides and \$15 a foot for the last side. The area of the pigpen must be 150 ft^2 . What should the dimensions be in order minimize the cost of the pigpen?

x is the two sides that cost \$10/ft

y is the side length where one side costs \$10/ft but the other costs \$15/ft

$$\text{Area: } xy = 150$$

$$\text{Solving for } y: y = \frac{150}{x}$$

$$\text{Cost } C = 10x + 10x + 10y + 15y$$

$$C = 20x + 25y$$

$$C(x) = 20x + 25\left(\frac{150}{x}\right)$$

$$C(x) = 20x + \frac{3750}{x}$$

$$C'(x) = 20 - \frac{3750}{x^2}$$

$$20 - \frac{3750}{x^2} = 0$$

$$20 = \frac{3750}{x^2}$$

$$20x^2 = 3750$$

$$x^2 = \frac{3750}{20} = 187.5$$

$$x \approx 13.69$$

Now find y :

$$y = \frac{150}{13.69}$$

$$y \approx 10.95$$

Dimensions: about 13.69 ft $\times$ 10.95 ft

3) Use the second derivative to find all intervals where $y = -x^4 + 5x^3 - x^2 + 4x + 1$ is concave up.

$$y' = -4x^3 + 15x^2 - 2x + 4$$

$$y'' = -12x^2 + 30x - 2$$

(concave up means $y'' > 0$)

$$-12x^2 + 30x - 2 > 0$$

find the zeros:

$$\left. \begin{aligned} -12x^2 + 30x - 2 &= 0 \\ 12x^2 - 30x + 2 &= 0 \\ 6x^2 - 15x + 1 &= 0 \end{aligned} \right\} X = \frac{15 \pm \sqrt{201}}{12}$$

and since y'' opens downward, it is positive between the two roots

$$\left(\frac{15 - \sqrt{201}}{12}, \frac{15 + \sqrt{201}}{12} \right)$$

$$\text{Approx: } (0.069, 2.431)$$

4) Find the smallest positive root of $\tan(x) = x + 1$ using Newton's Method.

Write down Newton's Equation and $x_0, x_1, x_2, \dots$

So my goal is the root of $\tan(x) = x + 1$

$$\text{Rewrite } f(x) = \tan(x) - x - 1$$

$$f'(x) = \sec^2(x) - 1$$

Newton's Method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

type 1 Enter (in radians)
↳ Ans = 1

$$\text{Ans} = \frac{\tan(\text{Ans}) - (\text{Ans}) - 1}{(\tan(\text{Ans}))^2}$$

$$x_{n+1} = x_n - \frac{\tan(x_n) - x_n - 1}{\sec^2(x_n) - 1}$$

Using $x_0 = 1$ ↴

$$x_1 = 1 - \frac{\tan(1) - 1 - 1}{\sec^2(1) - 1}$$

$$\approx 1.102473$$

$$x_1 \approx 1.102473$$

$$x_2 \approx 1.138635$$

$$x_3 \approx 1.132373$$

$$x_4 \approx 1.132268$$

5) Use the midpoints of 10 rectangles to estimate $\int_0^5 e^{-x} dx$. Either show the area of all 10 rectangles, or show how to use sum(sequence... on the calculator.

2nd STAT
Math
Sum(

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OPS
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ALPHA)
For N

$$\Delta x = \frac{5-0}{10} = \frac{5}{10} = 0.5 \quad (\text{width})$$

mid points

0.25 0.75 1.25 1.75 2.25 2.75 3.25 3.75 4.25 4.75

$$M_{10} = 0.5 (e^{-0.25} + e^{-0.75} + e^{-1.25} \dots \text{etc for all midpoints})$$

$$\text{Sum}(\text{Seq}(0.5 * e^{-(0.5n - 0.25)}, n, 1, 10))$$

$$\approx 0.983$$

6) A farmer has 20 cows that each produce 7 gallons of milk each day. Every time she gets 2 new cows, the milk production decreases by 1 gallon for each cow. How many cows should the farmer have to maximize total milk production? - everytime she adds cows it decreases production.

X is the number of groups of 2 new cows

So the total number of cows is: $20 + 2x$

Every group of 2 new cows makes each cow

Produce 1 less gallon, so gallons per cow is:

number of starting milk - number of times

she added 2 cows $7 - x$

Total milk production

$$\begin{aligned} P(x) &= (20 + 2x)(7 - x) \\ &= 140 - 20x + 14x - 2x^2 \\ &= 140 - 6x - 2x^2 \end{aligned}$$

$$P'(x) = -6 - 4x = 0$$

$$-4x = 6$$

$$x = -\frac{3}{2}$$

but she can't add negative cows

↳ She should keep what she has (20 cows) to maximize milk production